

(PROJECT PROPSAL)



Implement Model Predicting Diabetes , Hypertension and Dementia

**CHATCHOL CHAIWONGSRIAROON
THITIPONG MUNWONG
THANAPON POMSANAM
RATTIPHONG YEEPHINYO
NIPPIT CHATWARNITKUN**

**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2022

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

Implement Model Predicting Diabetes , Hypertension and Dementia

CHATCHOL CHAIWONGSRIAROON

THITIPONG MUNWONG

THANAPON POMSANAM

RATTIPHONG YEEPHINYO

NIPPIT CHATWARNITKUN

A COMPUTER ENGINEERING PROJECT SUBMITTED TO
MAE FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING

MAE FAH LUANG UNIVERSITY

2022

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

Implement Model Predicting Diabetes , Hypertension and Dementia

CHATCHOL CHAIWONGSRIAROON

THITIPONG MUNWONG

THANAPON POMSANAM

RATTIPHONG YEEPHINYO

NIPPIT CHATWARNITKUN

**THIS COMPUTER ENGINEERING PROJECT HAS BEEN
APPROVED**

**TO BE A PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE DEGREE OF BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

2022

EXAMINING COMMITTEE

ADVISOR.....
(Assoc. Prof. Dr.Punnarumol Temdee)

COMMITTEE.....
(Asst. Prof. Dr.Nattapol Aunsri)

COMMITTEE.....
(Asst. Prof. Gp.Capt. Dr.Thongchai Yooyativong)

© COPYRIGHT BY MAE FAH LUANG UNIVERSITY

ACKNOWLEDGEMENT

This project will not be successful without the kind support of the advisor, committees and all supportive persons. We would like to thank everyone for their advice and help. We are also grateful to our parents and friends for their continuous encouragement throughout this project. This accomplishment would not have been possible without them. Thank you.

CHATCHOL CHAIWONGSRIAROON

THITIPONG MUNWONG

THANAPON POMSANAM

RATTIPHONG YEEPHINYO

NIPPIT CHATWARNITKUN

Title	Model Predicting Diabetes , Hypertension and Dementia
Author	Mr.CHATCHOL CHAIWONGSRIAROON Mr.THITIPONG MUNWONG Mr.THANAPON POMSANAM Mr.RATTIPHONG YEEPHINYO Mr.NIPPIT CHATWARNITKUN
Degree	Bachelor of Engineering (Computer Engineering)

Supervisory Committee

Assoc. Prof. Dr.Punnarumol Temdee	Advisor
Asst. Prof. Dr.Nattapol Aunsri	Committee
Asst. Prof. Gp.Capt. Dr.Thongchai Yooyativong	Committee

Abstract

This senior project proposes to predict the patient that if they had a sickness or not , the sickness that we scope is the sickness that predict that will increase in a daytime such as the Diabetes , Hypertension and Dementia by themselves for help the doctor or the nurse from their time can cure the patient that have confirmed that they have a sickness and it also help the patient or the normal people that have anxiety to be the firstly check by themselves and help them to be safer than to go to the hospital for a check in this situation of epidemic of Covid-19.

The results showed that our prediction is complete as indicated in the scope. In addition, the survey of 20 persons revealed that our project could help normal people , doctors or nurses and the people who work in the medical field more or less.

Keyword: Diabetes , Hypertension , Dementia , Predict,

CONTENTS

ACKNOWLEDGEMENT	i
Abstract	ii
CONTENTS	iii
LIST OF TABLES	iv
LIST OF FIGURES	v
CHAPTER 1 INTRODUCTION	1
1.1 Background and Rationale	1
1.2 Objective	2
1.3 Plan	2
1.4 Methodology	2
1.5 Plan	3
1.6 Expected Result	3
1.7 Place and Equipment	3
CHAPTER 2 LITERATURE REVIEW	4
2.1 Related Theory	4
2.2 Related Research	5
2.3 Related Technology	6
CHAPTER 3 ANALYSIS AND DESIGN	7
3.1 Existing System	7
3.2 New System	7
3.3 Use case diagram	8
3.4 Activity diagram	9
CHAPTER 4 CONCLUSION	10
4.1 Conclusion	10
4.2 Discussion	10
REFERENCES	10

LIST OF TABLES

Table	Page
Table 1.1 Working plan	3

LIST OF FIGURES

Figure	Page
Figure 2.1 Colab logo	6
Figure 2.2 Anaconda logo	6
Figure 2.2 Excel logo	6
Figure 3.1 Use case diagram of the existing system	7
Figure 3.2 Use case diagram of the Model	8
Figure 3.3 Activity diagram of the new system	9

CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

Today this Today the covid-19 situation causes many diseases and it can be easily caused. Our medication lacks makes patients don't have a cure or found out too late and in some cases causes death. After the research from the hospital, we found that most of the medical personnel agree that our medical facility is lacking so we need to know the rate of incurring disease that will occur because we want to take good care of our patients, cure the disease efficiently and protect or decrease the death rate. In our group's opinion we want to help the medical facility. We will create a model that helps medical personnel to predict the incurrent disease. To create a model of prediction we have to analyze the data that we have from the hospital and Data Warehousing. The data will be in data science by using classification, neural networks, etc. after that, we will find the accuracy by using python or math Laps to find the highest accuracy.

Diabetes and Hypertension is common in our country and nowadays we have covid19 so It is hard to go to the hospital because of the risk of infection. So our group wants to help people that are concerned about their health so we are using data from people that have Diabetes or Hypertension to predict the people that may have this disease also.

Our group choose diabetes and hypertension to be the disease that our model going to predict because in thailand nowadays Those disease found easily and can cause death to patient if you found out to late. Our group doesn't want to predict the fact that the user have disease we want to predict the possibility that the user are going to have high risk to have those diseases. The way that we going to use is the medical personnel will bring us the information from the labs and put it in to our model. The model will predict the result and give it to the doctor to making the decision. The result will be 2 choice first is the patient have high risk to have a disease and the patient have low risk to have a disease.

1.2 Objective

1.2.1 We need to create the highest accuracy model to predict the possibility of patient that may have Diabetes with Hypertension and patient that have high risk to have Dementia.Scope

1.2.2 We will use python in the collapse.

1.2.3 Data from people that we receive by the patient that health check in the hospital.

1.3 Scope

1.3.1 General user

- 1) We can predict the predict the possibility of patient that may have Diabetes with Hypertension
- 2) We can predict the predict the possibility of patient that may have dementia

1.4 Methodology

1.2.4 Data collection (We use the data that our seniors receive from the hospital and then analyze this data for us.)

1.2.5 Model construction (We will use the data that we received then use data science to find the best way to create models such as decision trees ,neural net classificationetc.)

1.2.6 Model testing (We will find the highest accuracy for testing our example before testing in real life.)

1.5 Plan

Table 1.1 Working plan

	Pre-project					Project				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Create Topic										
Gathering Data										
Find Solution										
Manage Data										
Create Model										
Model testing										
Preparing Document										
Submit Model										

1.6 Expected Result

- 1.6.1 Have a model with high accuracy to prediction.
- 1.6.2 Everyone can use our model to predict The Diabetes, Hypertension and Dementia.

1.7 Place and Equipment

- 1.7.1 The place we get the information are the Hospital and Free souse.
- 1.7.2 Use python language
- 1.7.3 Use software Collapse.

CHAPTER 2

LITERATURE REVIEW

1.8 Related Theory

Diabetes and hypertension are common diseases that are usually found in Thailand but those diseases are very dangerous and can cause serious complications.[1] In 2014, 8.5% of adults aged 18 years and older had diabetes. In 2019, diabetes was the direct cause of 1.5 million deaths and 48% of all deaths due to diabetes occurred before the age of 70 years. So our group wants to use data mining such as Decision tree, K-Nearest Neighbor(KNN) and Random forest to predict the risk that a patient is going to have those diseases. Diabetes is a disease that occurs when cells in the patient's body are unfunctional by this symptom related to insulin and hormone that create pancreas. This disease can cause chest pain, heart attack, stroke and narrowing of arteries. It is very dangerous not only to the old people but also the people that lack of insulin or hormone.[2] Hypertension is the disease that the pressure in red blood vessels is higher than 140/90 that can cause many symptoms such as heart attack, kidney failure, myocardial ischemia and paralysis. But when the patient has those diseases at the same time it is called cardiovascular disease and diabetes. This symptom will occur when the patient has hypertension, high cholesterol and can't control the insulin in their body by this disease can cause 4 times of death rate when the patient has diabetes. Our group wants to predict the possibility of the patient that they might have another disease when they have diabetes or hypertension by using data mining.

Data mining technique used in this project is a technique used for classification and finding important features.[4] We use 3 classification for classified the data Decision tree, Extra tree and Random forest classifier after we have proposed a diabetes prediction model for better classification of diabetes when we find the important Classification accuracy is boosted with the new dataset compared to existing dataset and the diabetes prediction improving the accuracy of classification to create a model for the most accuracy and prediction of the disease. Then use 3 models to predict the disease KNN, decision tree and Regression. We use classification for classification, which is important because when we use the original data the result is error and low accuracy so we use classification to make data more accurate and usable for our model.

[5] Dementia is a very dangerous disease. This disease usually occurs in old ages. Patients normally have an abnormal cerebral cortex if this symptom occurs for a long time. Patients will have various symptoms followed by such as losing their mind or memory and making a wrong decision. Our groups wanted every family to have happy life in their old ages so we decided to make a model that can predict this disease before it is too late. [6] After the research we found that our diabetes and hypertension disease project is very similar to this project because it uses the same classifier such as decision tree and extra tree.

1.9 Related Research

[1] World health organization statistics on diabetes.

<https://www.who.int/en/news-room/fact-sheets/detail/diabetes>.

[2] Adam Felman.(2019). Everything you need to know about hypertension.

<https://www.medicalnewstoday.com/articles/150109>

[3] Azad, C., Bhushan, B., Sharma, R. et al. Prediction model using SMOTE, genetic algorithm and decision tree (PMSGD) for classification of diabetes mellitus. *Multimedia Systems* (2021).

<https://doi.org/10.1007/s00530-021-00817-2>

[4] Diabetes Prediction using Machine Learning Algorithms

Author links open overlay panel Aishwarya Mujumdar V V aidehi Dr. b

<https://doi.org/10.1016/j.procs.2020.01.047>

[5] ภาวะสมองเสื่อมในผู้สูงอายุ (Dementia) คีนาท แชนอก พยาบาลวิชาชีพชำนาญการ

<https://www.dmh.go.th/downloadportal/morbidity/dementia.pdf>

[6] Bharati, S., Podder, P., Thanh, D.N.H. et al. Dementia classification using MR imaging and clinical data with voting based machine learning models. *Multimed Tools Appl* (2022).

<https://doi.org/10.1007/s11042-022-12754-x>

1.10 Related Technology,

On Technology we use a software 3 software

1. Google Colaboratory we use colab main merge data, clean data and test my data. We use colab because it is a free resource to run a model or run code but the resource colab gives us depends on the packet it is free but it has a low resource.



Figure 2.1 Colab Logo

2. Anaconda jupyter is a good software to run python but it can't work with another friend. We use it to run more code if google colab can't run it.



Figure 2.1 Anaconda Logo

3. Microsoft Excel, After we have a file we use excel to look at my data to check if the solution of the data is correct or not.



Figure 2.1 Excel Logo

CHAPTER 3

ANALYSIS AND DESIGN

1.11 Existing System

The existing could be demonstrated by a use case diagram in Figure 3.1.

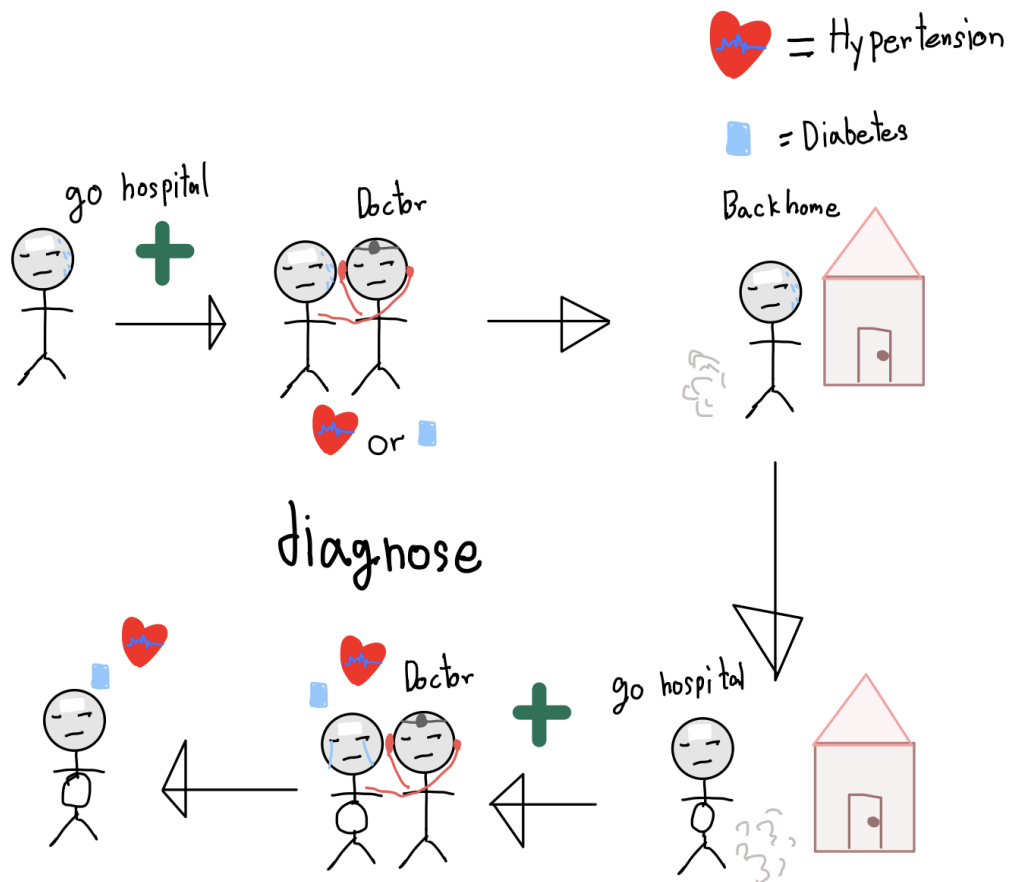


Figure 3.1 Use case diagram of the existing system

1.12 New System

Analyze and design the new system. Generally for a system, the analysis and design would include many diagrams such as Use case diagram, Activity Diagram, Flow chart, ER diagram, Data dictionary and User Interface design. Each of these topics could be a new subtopic (Heading 2).

1.13 Use case diagram

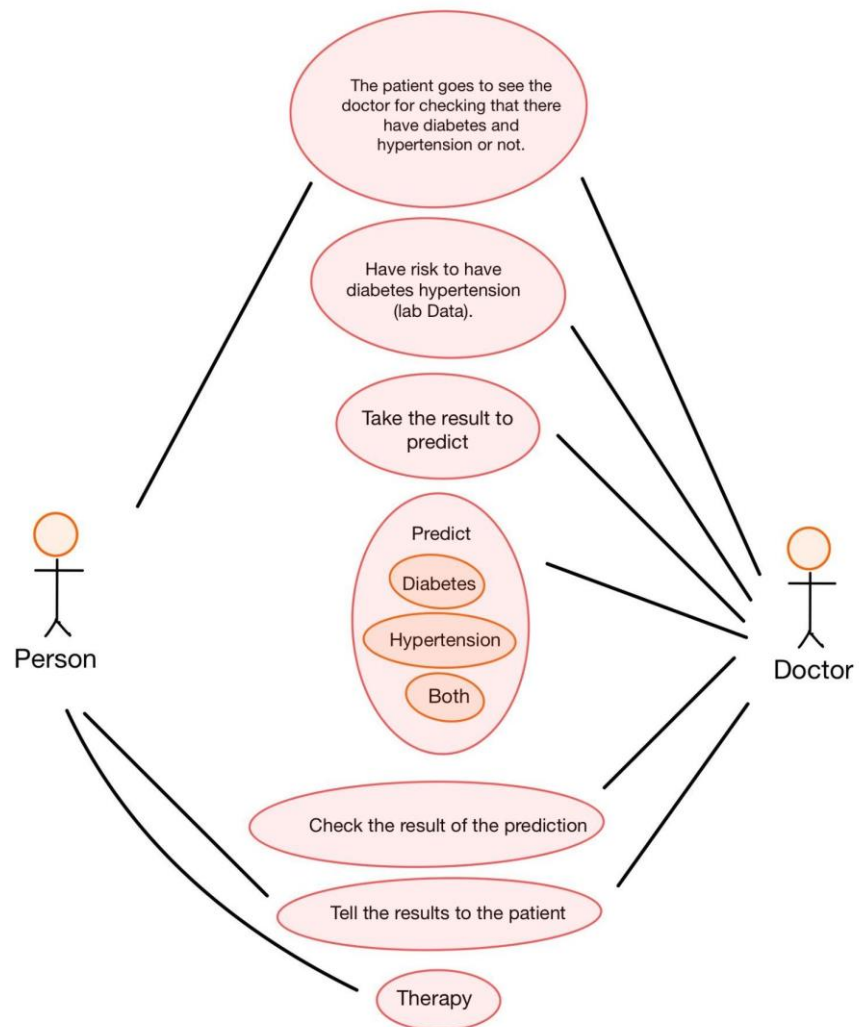


Figure 3.2 Use case diagram of the Model

1.14 Activity diagram

3.3. Analysis of a new system derives an activity diagram as shown in Figure

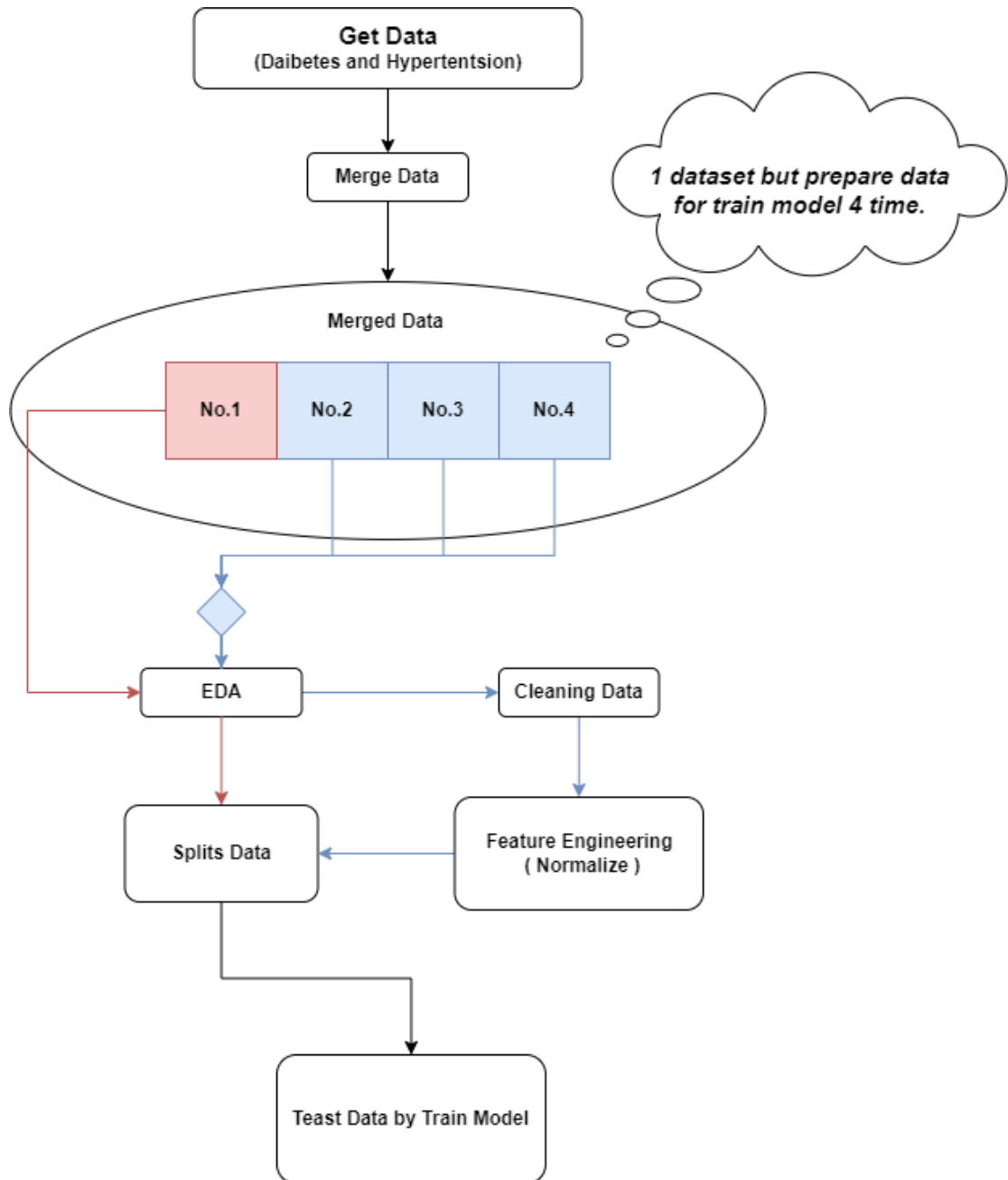


Figure 3.3 Activity diagram of the New system

REFERENCES

- [1] “Use case diagram - Wikipedia.” [Online]. Available: https://en.wikipedia.org/wiki/Use_case_diagram. [Accessed: 25-Apr-2017].
- [2] “Activity diagram - Wikipedia.” [Online]. Available: https://en.wikipedia.org/wiki/Activity_diagram. [Accessed: 25-Apr-2017].
- [3] “Kinect - Windows app development.” [Online]. Available: <https://developer.microsoft.com/en-us/windows/kinect>. [Accessed: 25-Apr-2017].
- [4] R. Miles, *Start Here! Learn the Kinect API*, 1 edition. Sebastopol, California: Microsoft Press, 2012.
- [5] “Processing.org.” [Online]. Available: <https://processing.org/>. [Accessed: 25-Apr-2017].
- [6] C. Reas and B. Fry, *Processing: A Programming Handbook for Visual Designers and Artists*, Second edition edition. Cambridge, Massachusetts: The MIT Press, 2014.

(Please use IEEE reference style)

2 CONCLUSION

2.5 Conclusion

Summarize what you have done in the senior project.

2.6 Discussion

Discuss the results. For example, explain your experimental results whether they are in accordance with the assumptions and give the reasons. In case of system development, you can discuss the problems and obstacles found during the entire processes including your solutions. Finally mention the suggestion for the further works.